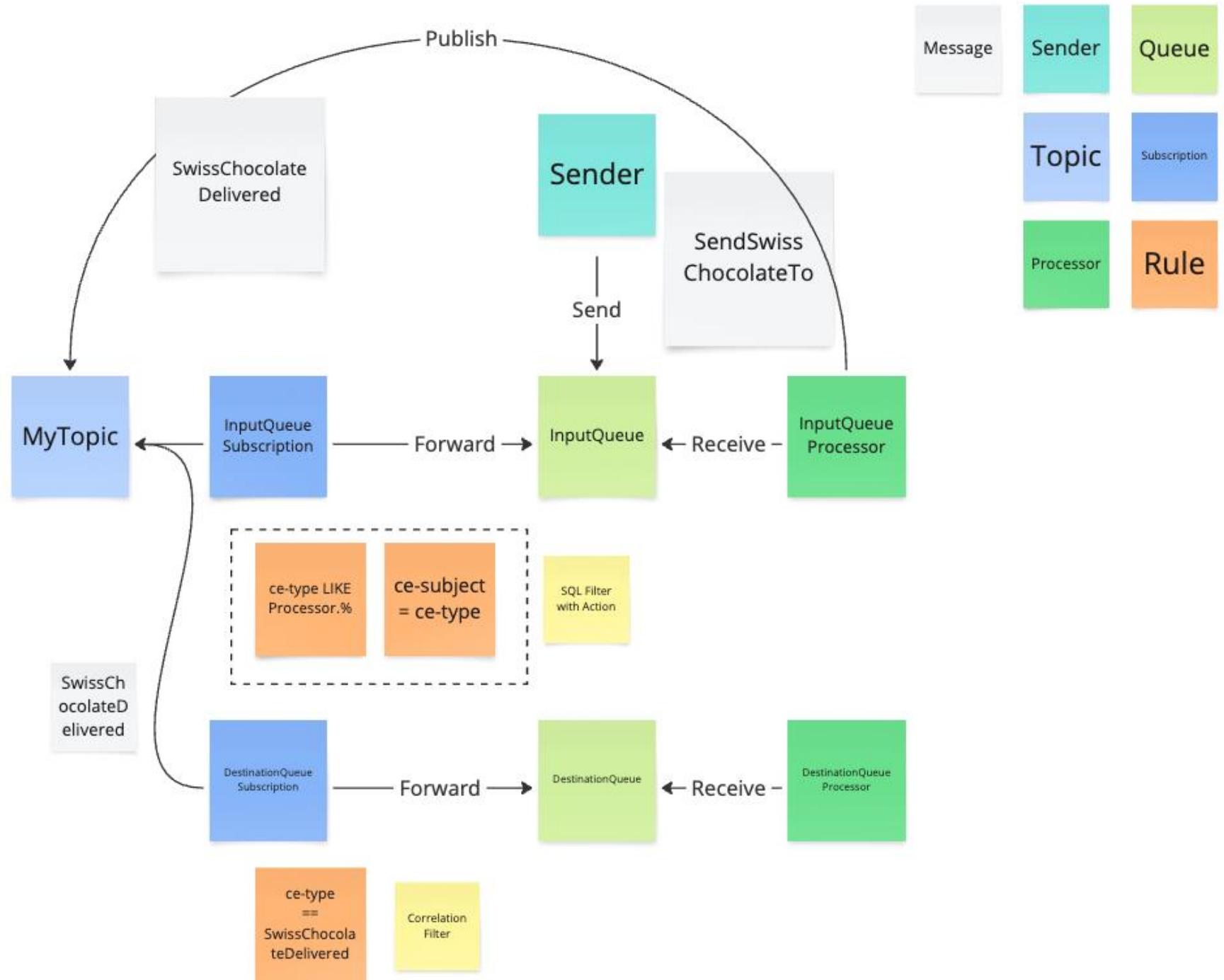


A row of various mailboxes in a grassy field with mountains in the background. The mailboxes are of different colors and styles, including white, orange, black, and rusty metal. Some have numbers like 1955, 405, 420, and 215. The background shows a clear blue sky and distant mountains.

Navigating through the **Azure Messaging** (over)choice

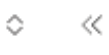
daniel.marbach@particular.net





swisschocolate~events (ndcosloservicebus1/swisschocolate~events) | Subscriptions

Service Bus Topic



Subscription

Refresh

Give feedback

- Overview
- Access control (IAM)
- Diagnose and solve problems
- Service Bus Explorer
- Resource visualizer
- Settings

Name	Status	Message count	Act
swisschocolate-delivery-tracking-subscription	Active	0	0
swisschocolatewarehouse-subscription	Active	0	0



Refresh



Delete



Give feedback



Overview



Diagnose and solve problems



Service Bus Explorer



Resource visualizer



Automation



Help

Filters

A conditional expression that's evaluated in the broker against the arriving messages' user-defined properties and system properties. [Learn more](#)

+ Add filter

Name

Filter Type

SwissChocolateDelivered

CorrelationFilter



SwissChocolateDelivered

Filter Type

SQL Filter

Correlation Filter

Filter messages using a set of conditions on system and user-defined properties that must all match. [Learn more](#)

SYSTEM PROPERTIES

Key

Value

CUSTOM PROPERTIES

Key

Value

ce-type

Processor.SwissChocolateDelivered



Save changes



swisschocolatewarehouse-subscription (ndcosloservicebus1/swisschocolate~events/swisschocolatewarehouse-subs...

Service Bus Subscription



Search



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Message Counts



Filters

A conditional expression that's evaluated in the broker against the arriving messages' user-defined properties and system properties. [Learn more](#)

+ Add filter

Name	Filter Type
AllEventsPublishedUnderNamespace	SqlFilter



AllEventsPublishedUnderNamespace

Filter Type

SQL Filter Correlation Filter

Filter messages by using a SQL-like expression that evaluates against message properties. [Learn more](#)

```
1 user.[ce-type] LIKE 'Processor.%'
```


CODE & DEMOS



- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Events
- Settings
- Geo-replication (preview)
- Shared access policies

Scale

- Geo-Recovery
- Networking
- Encryption
- Identity
- Configuration
- Properties
- Locks
- Entities
- Queues
- Topics
- Monitoring
- Automation
- Help

Configure

Run history

JSON

Notify

Diagnostic settings

Autoscale is a built-in feature that helps applications perform their best when demand changes. You can choose to scale your resource manually to a specific messaging unit, or via a custom Autoscale policy that scales based on metric(s) thresholds, or schedule messaging unit which scales during designated time windows. Autoscale enables your resource to be performant and cost effective by adding and removing instances based on demand. [Learn more about Azure Autoscale](#) or [view the how-to video](#).

Choose how to scale your resource

 **Manual scale**
Maintain a fixed messaging unit

 **Custom autoscale**
Scale on any schedule, based on any metrics

Custom autoscale

Autoscale setting name

danielpremium-Autoscale-713

Resource group

ReliableWorkshop

Messaging unit

2

Default *

Auto created default scale condition





Delete warning

 The very last or default recurrence rule cannot be deleted. Instead, you can disable autoscale to turn off autoscale.

Scale mode

☒ Scale based on a metric ☐ Scale to a specific messaging unit

Rules

 It is recommended to have at least one scale in rule. To create new rules, click [Add a rule](#)

Scale out

When	danielpremium	(Average) NamespaceCpuUsag...	Increase allowed next value
Or	danielpremium	(Average) NamespaceMemory...	Increase allowed next value

+ Add a rule

Messaging unit limits

Minimum *

1

Maximum *

16

Default *

1

Schedule

This scale condition is executed when none of the other scale condition(s) match



Metric source

Current resource (danielpremium)

Resource type

Service Bus Namespaces

Resource

danielpremium

Criteria

Metric namespace *

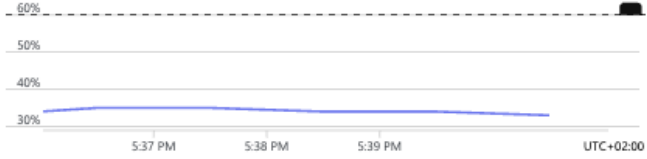
Standard metrics

Metric name

CPU

1 minute time grain

No dimension is available for the selected metric.



NamespaceCpuUsage (Average)

34 %

Operator *

Greater than

Metric threshold to trigger scale action *

60

%

Duration (minutes) *

5

Time grain (minutes)

1

Time grain statistic *

Average

Time aggregation *

Average

Action

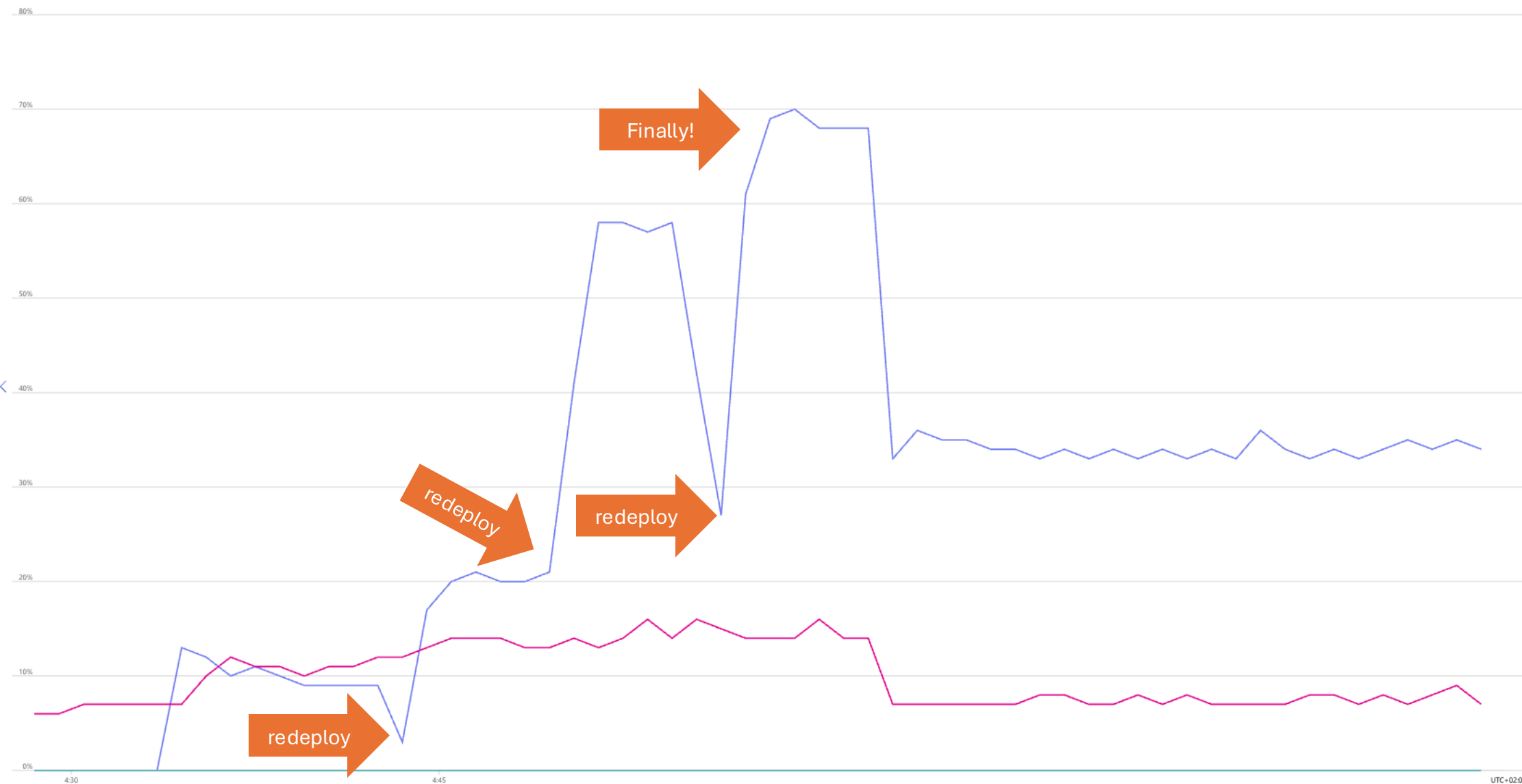
Operation *

Increase

Cool down (minutes) *

5

Current messaging unit is 2; Messaging unit can scale up and down among these values [1, 2, 4, 8, 16].

[danielpremium, CPU, Avg](#) [danielpremium, Memory Usage, Avg](#) [danielpremium, Throttled Requests., Sum](#)

CODE & DEMOS



Register an application ...

* Name

The user-facing display name for this application (this can be changed later).

ServiceBusRBAC ✓

Supported account types

Who can use this application or access this API?

- ☒ Accounts in this organizational directory only (Default Directory only - Single tenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)
- ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)
- ☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform ▼ e.g. https://example.com/auth

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

 ServiceBusRBAC | API permissions 

Refresh



Got feedback?



Overview



Quickstart



Integration assistant



Diagnose and solve problems



Manage



Branding & properties



Authentication



Certificates & secrets



Token configuration



API permissions



Expose an API



App roles



Owners



Roles and administrators



Manifest

> Support + Troubleshooting

Configured permissions

Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. [Learn more about permissions and consent](#)



Add a permission



Grant admin consent for Default Directory

API / Permissions name

Type

Description

▼ [Microsoft Graph \(1\)](#)[User.Read](#)

Delegated

Sign in and read user profile

▼ [Microsoft.ServiceBus \(1\)](#)[user_impersonation](#)

Delegated

Have full access to Azure Service Bus service

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, try



ServiceBusRBAC | Certificates & secrets

[Got feedback?](#)[Overview](#)[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)[Manage](#)[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)[Manifest](#)[Support + Troubleshooting](#)

Credentials enable confidential applications to identify themselves to the authentic scheme). For a higher level of assurance, we recommend using a certificate (instead

i Application registration certificates, secrets and federated credentials can be found

[Certificates \(0\)](#)[Client secrets \(1\)](#)[Federated credentials \(0\)](#)

A secret string that the application uses to prove its identity when requesting a to

[+ New client secret](#)

Description	Expires	Value ⓘ
ServiceBusClientSecret	8/14/2025	RXk8Q~v



ServiceBusRBAC

[Delete](#)[Endpoints](#)[Preview features](#)

Overview

[Quickstart](#)[Integration assistant](#)[Diagnose and solve problems](#)

▼ Manage

[Branding & properties](#)[Authentication](#)[Certificates & secrets](#)[Token configuration](#)[API permissions](#)[Expose an API](#)[App roles](#)[Owners](#)[Roles and administrators](#)[Manifest](#)[Support + Troubleshooting](#)

^ Essentials

Display name : [ServiceBusRBAC](#)

Application (client) ID : f68771ae-572

Object ID : 444d3304

Directory (tenant) ID : ffe0df17

Supported account types : [My organization only](#)

[Get Started](#)[Documentation](#)

The Microsoft id

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

🔍

Azure Er



Type : All

Category : All

Name ↑↓	Description ↑↓
Azure Service Bus Data Owner	Allows for full access to Azure Service Bus resources.
Azure Service Bus Data Receiver	Allows for receive access to Azure Service Bus resources.
Azure Service Bus Data Sender	Allows for send access to Azure Service Bus resources.
Log Analytics Reader	Log Analytics Reader can view and search all monitoring da
Monitoring Metrics Publisher	Enables publishing metrics against Azure resources

Showing 1 - 5 of 5 results.

Add role assignment ...

[Role](#)

[Members](#)

[Conditions](#)

[Review + assign](#)

Selected role

Azure Service Bus Data Sender

Assign access to

☒ User, group, or service principal

☐ Managed identity

Members

[+ Select members](#)

Name	Object ID
ServiceBusRBAC	01426532-

Description

Optional

~/Projects/AzureMessagingOverchoice/ServiceBus/ServiceBusRBAC git:(main)±17 (14.66s)

dotnet run -c Release

Using launch settings from /Users/danielmarbach/Projects/AzureMessagingOverchoice/ServiceBus/ServiceBusRBAC/Properties/launchSettings.json...

Unhandled exception. System.UnauthorizedAccessException: Unauthorized access. 'Listen' claim(s) are required to perform this operation. Resource: 'sb://ndcosloSERVICEBUS1.servicebus.windows.net/rbacqueue'. TrackingId:dba9c5a0083d48a0afc9e2b70fab27d8_G9, SystemTracker:gateway10, Timestamp:2025-05-16T16:01:25

For troubleshooting information, see <https://aka.ms/azsdk/net/servicebus/exceptions/troubleshoot>.

at Azure.Messaging.ServiceBus.Amqp.AmqpReceiver.ReceiveMessagesAsyncInternal(Int32 maxMessages, Nullable`1 maxWaitTime, TimeSpan timeout, CancellationToken cancellationToken)

at Azure.Messaging.ServiceBus.Amqp.AmqpReceiver.<>c.<<ReceiveMessagesAsync>b__44_0>d.MoveNext()

--- End of stack trace from previous location ---

at Azure.Messaging.ServiceBus.ServiceBusRetryPolicy.RunOperation[T1,TResult](Func`4 operation, T1 t1, TransportConnectionScope scope, CancellationToken cancellationToken, Boolean logTimeoutRetriesAsVerbose)





rbacqueue (ndcosloservicebus1/rbacqueue)

Service Bus Queue



Refresh



Delete



Give feedback



Overview



Access control (IAM)



Diagnose and solve problems



Service Bus Explorer



Resource visualizer



Settings



Automation



Help

^ Essentials

Namespace : [ndcosloservicebus1](#)

Status : [Active](#)

Partitioning : Disabled

Duplicate detection : Disabled

Dead lettering : [Disabled on message expiration](#)

^ Settings

Max delivery count
10 [\(change\)](#)

Current size
0.2 KB

Max size
1 GB [\(change\)](#)

Message time to live
14 DAYS [\(change\)](#)

^ Message Counts

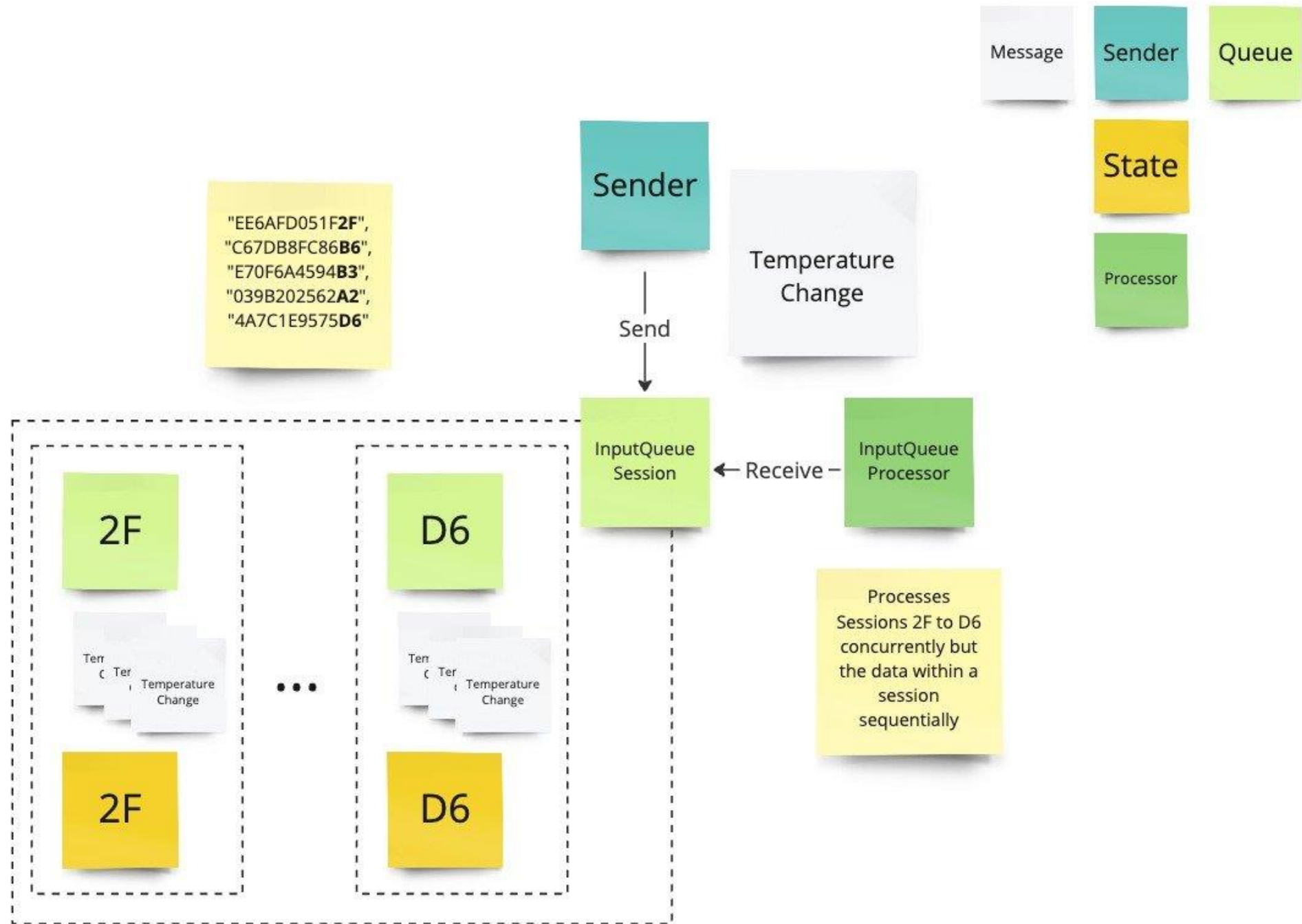
Active
1 MESSAGES

Scheduled
0 MESSAGES

Dead-letter
0 MESSAGES

Transfer
0 MESSAGES

Transfer d
0 MESSAGES





chocolatestorage~temperature (ndcosloservicebus1/chocolatestorage~temperature)

Service Bus Queue



Refresh | Delete | Give feedback

Auto refresh ☒ On Refresh after 10 seconds

Overview

- Access control (IAM)
- Diagnose and solve problems
- Service Bus Explorer
- Resource visualizer
- > Settings
- > Automation
- > Help

Essentials

Namespace	: ndcosloservicebus1	Queue URL	: https://ndcosloservicebus1.servicebus.windows.net/chocolatestorage/te...
Status	: Active	Created	: Friday, May 16, 2025 at 17:26:18 GMT+2
Partitioning	: Disabled	Updated	: Friday, May 16, 2025 at 17:26:18 GMT+2
Duplicate detection	: Disabled	Sessions	: Enabled
Dead lettering	: Disabled on message expiration	Forward messages to	: Disabled



Settings

Max delivery count	Current size	Max size	Message time to live	Auto-delete	Message lock duration	Free space
10 (change)	165.4 KB	1 GB (change)	UNBOUNDED (change)	NEVER (change)	5 SECONDS (change)	100.0 %

Message Counts

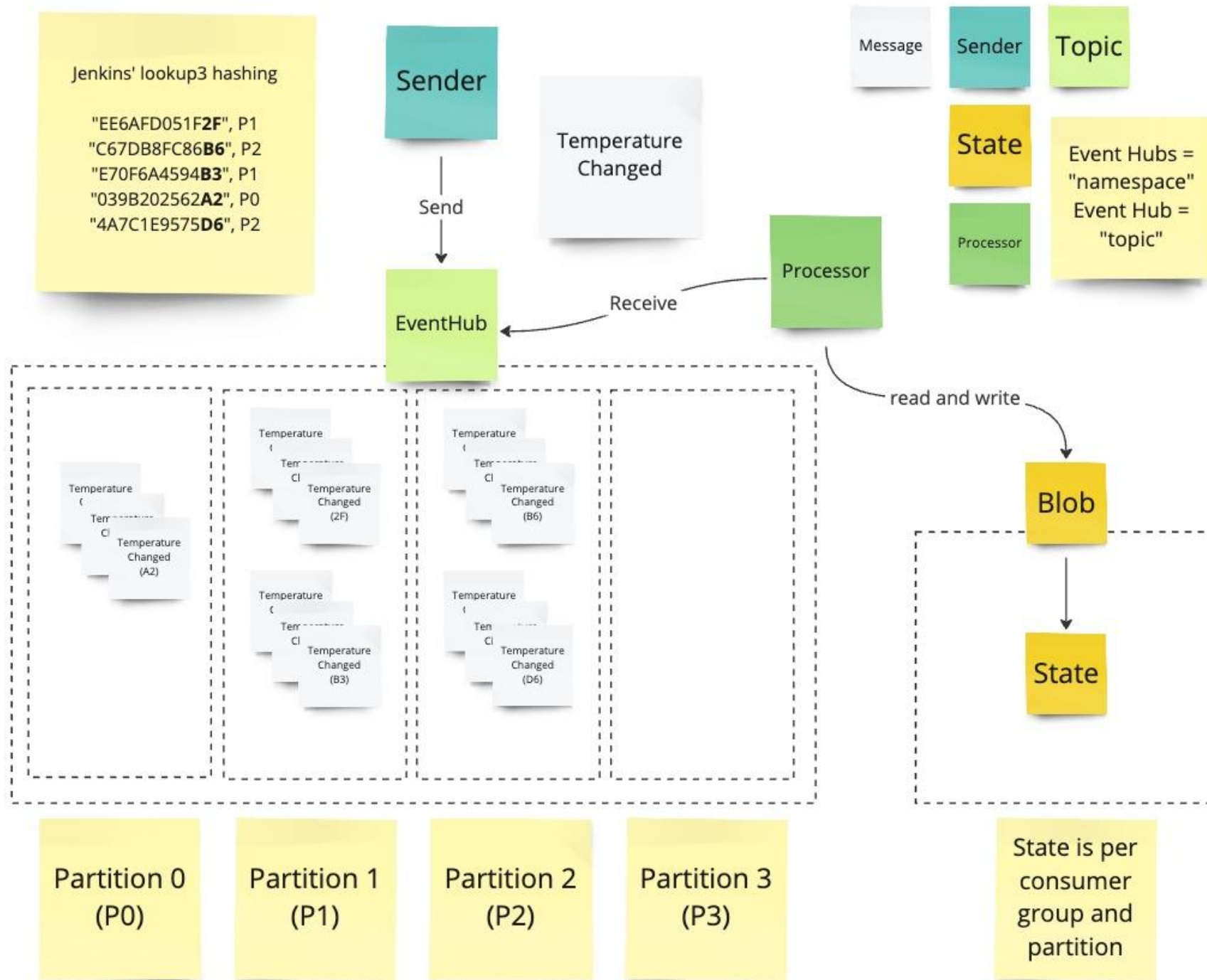
Active	Scheduled	Dead-letter	Transfer	Transfer dead-letter
259 MESSAGES	0 MESSAGES	0 MESSAGES	0 MESSAGES	0 MESSAGES

Metrics

Show data for the last: 1 hour 6 hours 12 hours 1 day 7 days 30 days

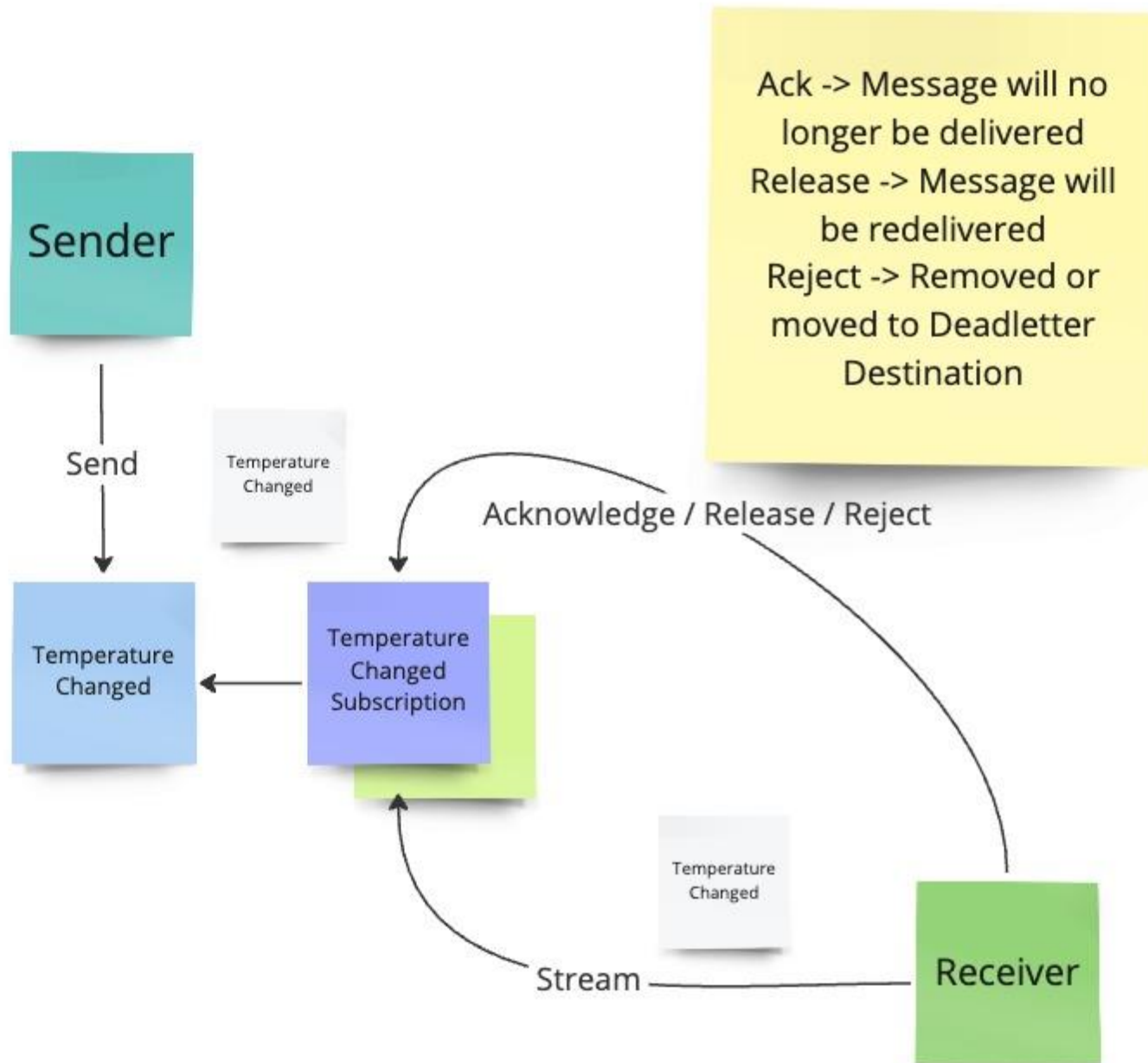
Requests

Messages



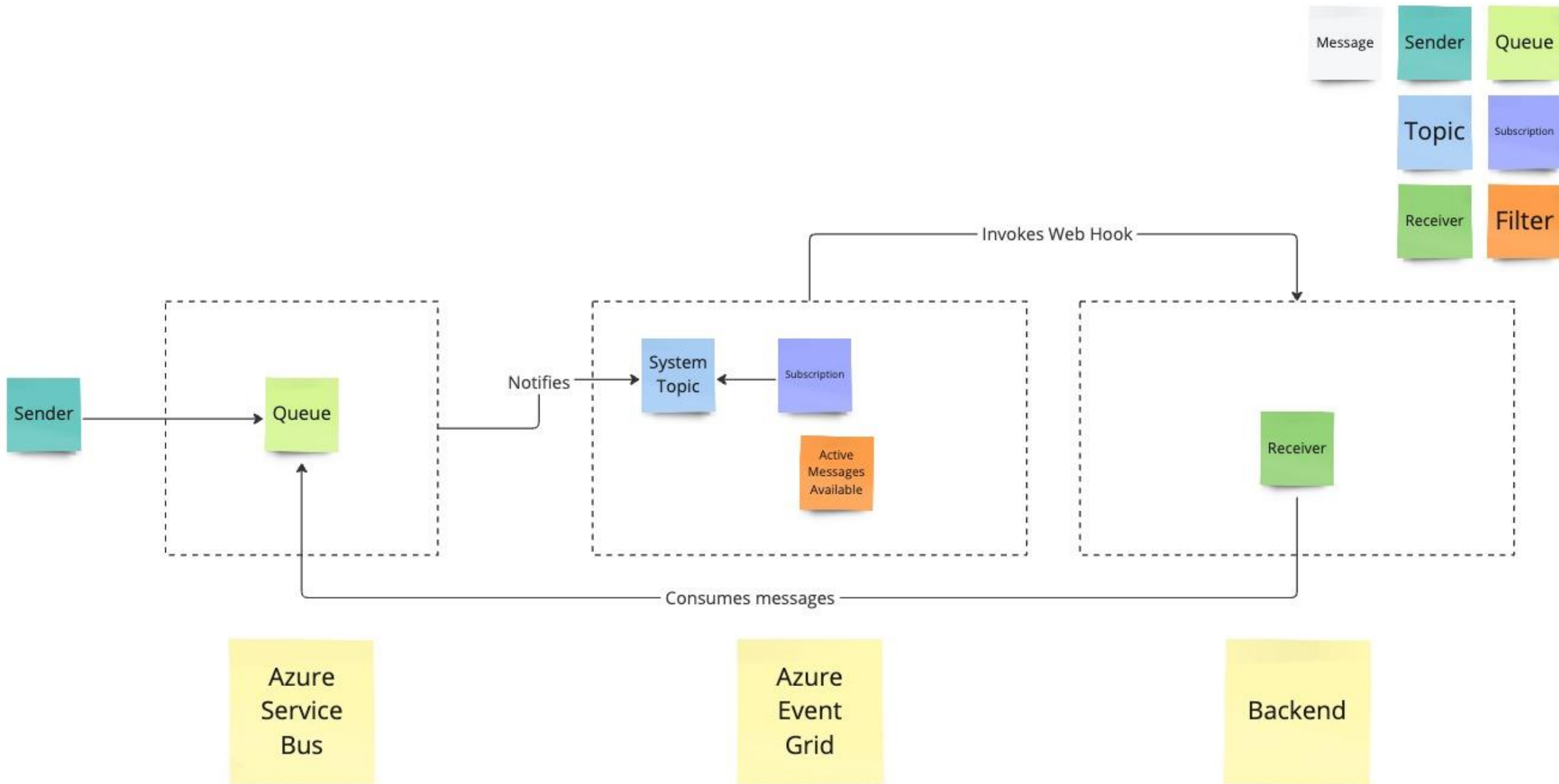
CODE & DEMOS





CODE & DEMOS





CODE & DEMOS



Segment	Eventing PubSub	Streaming	Enterprise Messaging
Product	Event Grid	Event Hubs	Service Bus
What do you care about	• Communication between apps / orgs • Individual message • Push and pull semantics • Filtering and routing • Pay as you go • Fan out	• Many messages in a Stream (think in MBs) • Ease of use and operation • Low cost • Fan in • Strict ordering • Kafka compliant • Works with other tools	• Instantaneous consistency • Strict ordering • JMS • Non-repudiation & Security • Geo-Replication & Availability • Rich features (de-dupe, scheduling, etc.)
What you care less about	• Ordering of messaging • Instantaneous consistency	• Individual message semantics • Server-side cursor • Only Once	• Cost • Simplicity
Focus point	• Reliable fan out push at massive scale	• High scale distributed log	• Highly indexed full featured message broker
	Serverless	Streaming Data	Enterprise



Use all of them



A row of various mailboxes in a grassy field with mountains in the background. The mailboxes are of different colors and styles, including white, orange, black, and rusty metal. Some have numbers like 1955, 405, 420, and 215. The background shows a clear blue sky and distant mountains.

Navigating through the **Azure Messaging** (over)choice

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